INCH-POUND

MIL-S-51422B(EA)
AMENDMENT 5
9 May 1991
SUPERSEDING
AMENDMENT 4
28 September 1990

MILITARY SPECIFICATION

SHELTER AND ENTRANCE ASSEMBLY - CHEMICAL-BIOLOGICAL SHELTER SYSTEM: INFLATABLE

This amendment forms a part of MIL-S-51422B(EA), dated 24 September 1987, and is approved for use by the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1.1: Under FEDERAL SPECIFICATIONS add the following:

"L-P-1040 - Plastic Sheets and Strips (Polyvinylfluoride)."

PAGE 2

2.1.2: Under DRAWINGS add the following:

"5-19-11247 - Nylon/Polyethylene Film Laminate.
QAP5-19-11247 - Nylon/Polyethylene Film Laminate."

AMSC N/A

FSC 4240

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PAGE 3

3.1.1: Add the following sentence to the end of the paragraph:

"For fabrics conforming to MIL-C-43944, the use of liquid mustard (H) is not required in conjunction with the decontamination procedure."

PAGE 4

3.1.3: Delete in its entirety and substitute the following:

"**3.1.3 Window material.** The material used in the windows shall be made out of either polyvinylfluoride laminate sheet or out of nylon/polyethylene film laminate as specified in either 3.1.3.1 or 3.1.3.2.

3.1.3.1 Polyvinylfluoride. This window material shall be a homogeneous polyvinylfluoride laminate sheet 0.008 + 0.001 inches thick formed by heat and pressure from a minimum of two sheets of material conforming to the requirements of type I, transparent, grade C, class 1 of L-P-1040 except for the following:

(1) The haze requirement for the single sheet is eliminated.
(2) The material is not limited to the 0.001 inch thickness.

In addition the window material shall conform to the following requirements:

(a) After accelerated weathering in accordance with Method 5804 Fed. Test Method Std. No. 191 for 200 hours, the window material shall have a haze factor not exceeding 25 when tested as specified in 4.4.11.

(b) The window material shall maintain flexibility and there shall be no indication of cracking of delamination at -65°F when tested as specified in 4.4.10.2.

(c) The initial and after low temperature agents penetration resistance shall be no less than 100 minutes for Mustard (H) and 200 minutes for GB when tested as specified in 4.4.10.

3.1.3.2 Nylon/polyethylene. This window material shall be made out of nylon/polyethylene film laminate conforming to the requirements on drawing 5-19-11247 except for the following:

(1) The material does not have to be capable of being bonded by heat electric

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or extrusion sealing devices.

(2) The finished width requirement is eliminated.

(3) The resistance to toxicological agent requirements shall be replaced with the following:

> The initial and after low temperature agents penetration resistance shall be no less than 100 minutes for Mustard (H) and 200 minutes for GB when tested as specified in 4.4.10.

(4) The cold crack test shall not be performed.

In addition the window material shall conform to the test specified in 4.4.12."

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3.1.8, line 2: Delete "(see 6.5)".

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3.3.5.4: Change from **"Scratch or abrastions"** to **"Scratch or abrasions"**.

PAGE 10

4.2: Delete in its entirety and substitute the following:

"**4.2 Preproduction inspection.** A preproduction sample of four shelter and entrance assemblies submitted in accordance with 3.4 shall be examined and tested for all requirements of the applicable drawings and this specification (see 6.2)."

4.3(b), line 3: Delete "polyurethane film" and substitute "polyvinylfluoride laminate film or nylon/polyethylene laminate film".

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4.4.10.2:

Line 4: Insert the following sentence after the words "...shall not be determined.":

"A second 4 inches by 4 inches specimen shall be exposed for 60 ± 10 minutes at

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a temperature of -25°F ± 1.8°F and tested as specified in Method 5874 of Fed. Test Method Std. 191 except that the hydrostatic resistance shall not be determined."

Lines 4 and 5:

Delete "Samples" and substitute "Both samples".

Delete "(if laminated)".

4.4.10.3, line 3: Delete "test" and substitute "tests".

4.4.11, line 3: Delete "3.1.3" and substitute "3.1.3.1".

4.4.12: Add the following new paragraph:

"**4.4.12 Nylon/polyethylene window material.** The nylon/polyethylene laminate film shall be tested in accordance with QAP5-19-11247, with the exception of cold crack, and shall conform to paragraph 3.1.3.2."

PAGE 24

6.2(d): Delete in its entirety and substitute the following:

"(d) When and for what purpose any of the following will be provided as Government furnished equipment (GFE):

Inflatable shelter	-	NSN 4240-00-152-1304
Inflatable entrance	-	NSN 4240-00-152-1306
Filter and utilities unit M68	-	NSN 4240-00-152-1305
Shelter system M51	-	NSN 4240-00-854-4144."

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6.7: Insert "GB (sarin)" between "Chemical-biological shelter system" and "Inflatable entrance" and insert "Mustard" between "Inflatable shelter" and "Protective shelter".

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